Pure Mathematics 2 · Original adaptations · 2026

Chapter 5
Sequences and series

数列与级数 · 30 questions with Chinese solutions

Answer all parts. Show sufficient working.
Give exact answers unless a degree of accuracy is requested.
Diagrams are not necessarily drawn to scale.

考法参照2024年前历年真题，题目与配图实质改编。
本册不是Pearson原卷；建议评分点非官方评分标准。
第8版 · 内部检查稿，未完成外部教师逐题审阅。

An arithmetic sequence has terms u_n and sum S_n. It is given that $S_4 = 3u_5$ and $u_9 = 42$.

(a) Show that the first term is six times the common difference. (2)

(b) Find a and d. (2)

(c) Find the least integer n for which $S_n > 2000$. (3)

Total for Question 1: 7 marks

A sequence has $S_n = 3n^2 + 5n$ for every positive integer n, where S_n is the sum of its first n terms.

(a) Find u_n and show that the sequence is arithmetic. (2)

(b) Express $S_{2n} - 2S_n$ in its simplest form. (2)

(c) The sum of terms u_{n+1} to u_{2n} is 950. Find n. (3)

Total for Question 2: 7 marks

For an arithmetic series, $S_{10} = 215$ and $S_{20} = 730$.

(a) Find the common difference. (2)

(b) Find the first term. (2)

(c) Find u_{15}. (2)

(d) Find the sum of the terms from u_{21} to u_{30}. (2)

Total for Question 3: 8 marks

The terms of an arithmetic sequence satisfy $u_3 = 11$ and $u_8 = 31$. A new sequence is formed by selecting $u_2, u_5, u_8, \ldots$.

(a) Find the first term and common difference of the original sequence. (2)

(b) Find a formula for the nth term of the new sequence. (2)

(c) Find the sum of all selected terms whose original index is at most 50. (3)

Total for Question 4: 7 marks

A convergent geometric series has sum to infinity 54. The sum of its first three terms is 38. Its common ratio is real.

(a) Show that the common ratio satisfies $r^3 = \frac{8}{27}$. (2)

(b) Find the first term. (2)

(c) Find the least number of terms needed for the remaining sum to be less than 1. (3)

Total for Question 5: 7 marks

A geometric sequence has positive terms. The sum of its second and third terms is 30, and the sum of its third and fourth terms is 15.

(a) Find the common ratio. (2)

(b) Using your common ratio, find

(i) the first term, (2)

(ii) the sum to infinity. (2)

(c) Find the sum of the terms with even indices. (3)

Total for Question 6: 9 marks

A geometric series has first term 4 and common ratio $-\frac{1}{2}$. Its first n terms have sum S_n.

(a) Find S_∞ and express S_n in terms of n. (2)

(b) State whether S_n is greater than or less than S_∞ when n is odd. Justify your answer. (2)

(c) Find the least n for which $|S_n - S_\infty| < 0.01$. (3)

Total for Question 7: 7 marks

An arithmetic sequence has non-zero common difference d. Its second, fourth and ninth terms, in that order, form a geometric sequence. Its fifth term is 13.

(a) Show that $d = 3a$, where a is the first term. (3)

(b) Find the common ratio of the geometric sequence. (2)

(c) Find the sum of the first 12 terms of the arithmetic sequence. (2)

Total for Question 8: 7 marks

A sequence is defined by $u_1 = 3$ and $u_{n+1} = 12 - u_n$ for $n \geq 1$.

(a) Find u_2, u_3, u_4. (2)

(b) Using the repeating pattern, find

(i) u_{101}, (2)

(ii) S_{101}. (2)

(c) A student claims that u_n tends to 6. Explain why the claim is false. (2)

Total for Question 9: 8 marks

A sequence satisfies $u_1 = 2$ and $u_{n+1} = 3u_n + 4$. Define $v_n = u_n + 2$.

(a) Show that $v_{n+1} = 3v_n$. (2)

(b) Find an expression for u_n. (2)

(c) Show that $S_n = 2(3^n - 1) - 2n$. (3)

Total for Question 10: 7 marks

A tank contains 10 litres immediately after the first treatment. Each further treatment removes 20% of the liquid and then adds 6 litres. Let u_n be the volume immediately after treatment n.

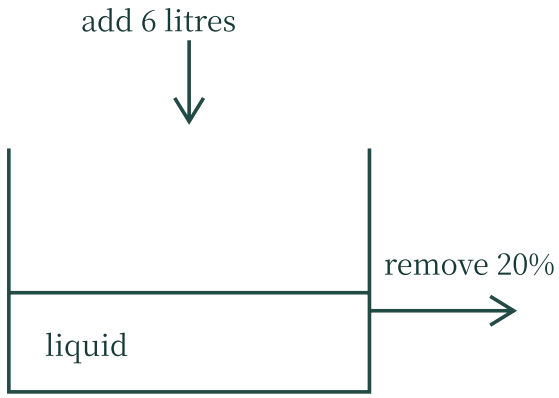


Diagram not necessarily drawn to scale

(a) Write down a recurrence relation for u_n. (2)

(b) Given that $u_n = 30 - 20(0.8)^{n-1}$, find the limiting volume. (2)

(c) Find the first treatment after which the volume is at least 29 litres. (3)

Total for Question 11: 7 marks

A saver deposits £40 in week 1, increasing the weekly deposit by £5 until week 8. From week 9 onwards, each weekly deposit is 90% of the previous week's deposit.

(a) Find the total deposited in the first 8 weeks. (2)

(b) Find the total deposited in weeks 9 to 16 inclusive. (3)

(c) Find the total amount deposited if the scheme continues indefinitely. (2)

Total for Question 12: 7 marks

In plan A a runner covers 2 km in week 1 and 0.4 km more each subsequent week. In plan B the runner covers 2 km in week 1 and 10% more each subsequent week.

(a) Find the distance in week 6 under

(i) plan A, (2)

(ii) plan B. (2)

(b) Find the total distance in the first 10 weeks under each plan. (3)

(c) Find the first week in which plan B exceeds 6 km. (2)

Total for Question 13: 9 marks

An arithmetic sequence has first term a and common difference $d > 0$. The sum of its first n terms is 150, and the sum of its next n terms is 350.

(a) Show that $n^2 d = 200$. (2)

(b) Given $a = 6$, show that $3n^2 - 25n - 50 = 0$. (3)

(c) Find n and d. (2)

Total for Question 14: 7 marks

A convergent geometric series has positive first term a and common ratio r. The sum of all odd-indexed terms is 12 and the sum of all even-indexed terms is -4.

(a) Find r. (2)

(b) For the geometric series determined in part (a), find

(i) the first term, (2)

(ii) the sum of its first four terms. (3)

Total for Question 15: 7 marks

A display is built in rows of square tiles. The first row has 7 tiles and each subsequent row has 4 more tiles. The diagram shows the first three rows.



Diagram not necessarily drawn to scale

(a) Find an expression for the number of tiles in row n. (2)

(b) Show that the first n rows contain $n(2n + 5)$ tiles. (2)

(c) A builder has 418 tiles. Find the greatest number of complete rows and the number of unused tiles. (3)

Total for Question 16: 7 marks

A ball is dropped from a height of 5 m. After each impact it rises to 60% of the height from which it last fell. Air resistance during each flight is ignored.

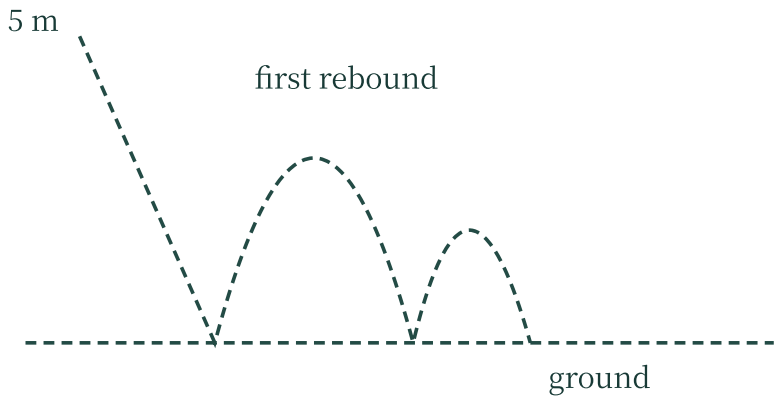


Diagram not necessarily drawn to scale

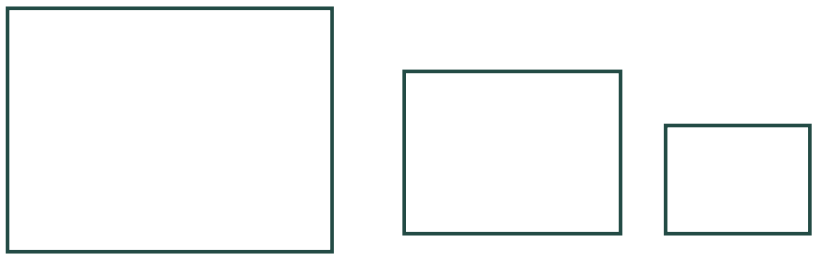
(a) Find the height of the third rebound. (2)

(b) Find the total distance travelled up to the second impact with the ground. (2)

(c) Find the limiting total distance travelled. (3)

Total for Question 17: 7 marks

A sequence of rectangles is formed by reducing each side length by a factor of $\frac{2}{3}$ at each step. The first rectangle has area $81\ \text{cm}^2$.



rectangle 1 rectangle 2 rectangle 3

Diagram not necessarily drawn to scale

(a) Find the area of rectangle n. (2)

(b) Find the sum of the areas of all the rectangles. (2)

(c) Find the first rectangle whose area is less than $1\ \text{cm}^2$. (3)

Total for Question 18: 7 marks

A sequence is given by $u_n = 2n + 1 + 4(-1)^{n+1}$.

(a) Find u_1, u_2, u_3. (2)

(b) Show that $S_{2m} = 4m(m+1)$. (3)

(c) Find the sum of terms u_{11} to u_{30}. (2)

Total for Question 19: 7 marks

Let $u_n = 1 + \sin^2\left(\frac{n\pi}{2}\right)$ for positive integers n.

(a) Find the first four terms. (2)

(b) Find $\sum_{n=1}^{75} u_n$. (2)

(c) Find $\sum_{n=1}^{75} (3n - u_n)$. (3)

Total for Question 20: 7 marks

A sequence is defined by $u_1 = 2$ and $u_{n+1} = 1 - \dfrac{1}{u_n}$.

(a) Find u_2, u_3, u_4. (2)

(b) Explain why the sequence has period 3, and find u_{100}. (2)

(c) Find S_{100}. (3)

Total for Question 21: 7 marks

The sum of the first n terms of a sequence is $S_n = 4(2^n - 1) + 3n$.

(a) Find u_1. (2)

(b) Show that $u_n = 2^{n+1} + 3$ for all $n \geq 1$. (3)

(c) Show that $u_{n+1} = 2u_n - 3$. (2)

Total for Question 22: 7 marks

A workshop makes 120 components in its first month. Monthly output then increases by 5% each month, according to a model which ignores rounding.

(a) Write an expression for the total output in the first n months. (2)

(b) Find the least n for which total output exceeds 2000. (3)

(c) The workshop closes after month 12. Find the proportion of its total output made in the last three months. (2)

Total for Question 23: 7 marks

A grant pays £900 in the first year and £60 less each following year. Payments stop before the first non-positive payment.

(a) Find the number of payments. (2)

(b) Find the total paid. (2)

(c) Find the least number of years needed to receive at least 75% of the total. (3)

Total for Question 24: 7 marks

A geometric sequence has first term 6 and common ratio $r > 1$. Its sums satisfy $S_{n+1} - S_n = 192$ and $S_n - S_{n-1} = 96$.

(a) Find r. (2)

(b) Find n. (2)

(c) Find $S_{2n} - S_n$. (3)

Total for Question 25: 7 marks

Let $S_n = a + ar + ar^2 + \cdots + ar^{n-1}$, where $r \neq 1$.

(a) By comparing S_n with rS_n, prove that $S_n = \dfrac{a(1 - r^n)}{1 - r}$. (3)

(b) A geometric series has $a = 9$ and $r = \dfrac{1}{3}$. Find

(i) the sum from the third to the sixth term inclusive, (2)

(ii) the sum of all terms after the sixth term. (2)

Total for Question 26: 7 marks

Three positive numbers in arithmetic progression have sum 30 and the sum of their squares is 332.

(a) Write the numbers as $10 - d, 10, 10 + d$, and show that $d^2 = 16$. (2)

(b) List the three numbers in increasing order. (2)

(c) The smallest and largest numbers are the first two terms of a geometric sequence. Find its fifth term. (3)

Total for Question 27: 7 marks

A machine uses 8 units of energy in each odd-numbered cycle and 5 units in each even-numbered cycle. A start-up charge of 20 units is incurred once.

(a) Find the energy used, including start-up, in the first 25 cycles. (2)

(b) Express the total for the first $2n$ cycles in terms of n. (2)

(c) A limit of 500 units includes start-up. Find the greatest number of complete cycles possible. (3)

Total for Question 28: 7 marks

A convergent geometric series has sum to infinity 96 and real non-zero common ratio r. For a positive integer N, the sum of all terms after term N is 12, and the sum of all terms after term $N+2$ is 3.

(a) Show that $r = \pm\frac{1}{2}$. (2)

(b) Determine the sign of r and the value of N, explaining why the other sign is impossible. (3)

(c) Find the first term and the sum of terms from $N+1$ to $2N$ inclusive. (3)

Total for Question 29: 8 marks

The points (n, u_n) lie on the straight line through $(2, 9)$ and $(7, 24)$, where n is a positive integer. Let $S_n = \sum_{r=1}^{n} u_r$.

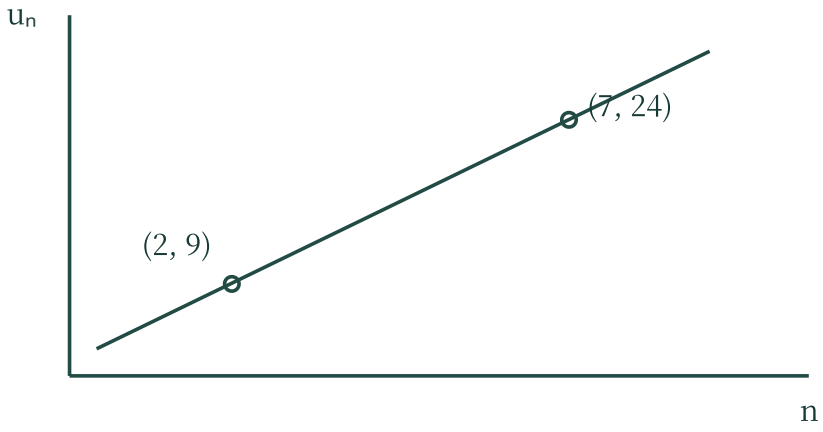


Diagram not necessarily drawn to scale

(a) Find u_n. (2)

(b) Show that $S_n = \frac{3}{2}n(n + 3)$. (2)

(c) Find the value of n for which $S_{n+1} - S_n = 60$. (2)

Total for Question 30: 6 marks

Answers and Suggested Mark Schemes

中文答案、解析与原创建议评分点

Answer 1

P2-C05-R01

原创改编（2026）· 主要考点：等差条件与阈值

考法参照：P2-2020-01-Q08（2020）：等差级数证明与交错项求和

由项与部分和关系确定参数，继而用单调部分和判断阈值；不保留原卷交错通项。

(a) · 2 marks

$a = 6d.$

$4a + 6d = 3(a + 4d)$，整理得$a = 6d$。

M1	$4a + 6d = 3(a + 4d)$，整理得$a = 6d$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$a = 18,\ d = 3.$

$u_9 = a + 8d = 14d = 42$，所以$d = 3$，$a = 18$。

M1	$u_9 = a + 8d = 14d = 42$，所以$d = 3$，$a = 18$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$n = 32.$

$S_n = \dfrac{3}{2}n(n + 11)$；$S_{31} = 1953$，$S_{32} = 2064$，因各项为正，最小值是32。

M1	$S_n = \dfrac{3}{2}n(n + 11)$；$S_{31} = 1953$，$S_{32} = 2064$，因各项为正，最小值是32。
A2	结论、定义域、精度与所需单位正确

Answer 2

原创改编（2026）· 主要考点：部分和反推通项
考法参照：P2-2020-01-Q08（2020）：等差级数证明与交错项求和
将求和公式证明改为部分和反推、区段和与二次方程。

(a) · 2 marks

$u_n = 6n + 2$，公差为6。

$u_n = S_n - S_{n-1} = 6n + 2$；$u_1 = S_1 = 8$也符合。

M1	$u_n = S_n - S_{n-1} = 6n + 2$；$u_1 = S_1 = 8$也符合。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$6n^2$.

$S_{2n} = 12n^2 + 10n$，减去$6n^2 + 10n$。

M1	$S_{2n} = 12n^2 + 10n$，减去$6n^2 + 10n$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$n = 10$.

$S_{2n} - S_n = 9n^2 + 5n = 950$，$(n - 10)(9n + 95) = 0$；舍去负根，得$n = 10$。

M1	$S_{2n} - S_n = 9n^2 + 5n = 950$，$(n - 10)(9n + 95) = 0$；舍去负根，得$n = 10$。
A2	结论、定义域、精度与所需单位正确

Answer 3

P2-C05-R03

原创改编（2026）· 主要考点：等差区段和
考法参照：P2-2020-10-Q05（2020）：两种等差储蓄方案的合理根
改为两段累计和辨识数列，再计算非首项区间。

(a) · 2 marks

$d = 3$.

用前20项的和减去两倍前10项的和，可以消去首项：$S_{20} - 2S_{10} = 100d = 300$。

M1	用前20项的和减去两倍前10项的和，可以消去首项：$S_{20} - 2S_{10} = 100d = 300$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$a = 8$.

$10a + 45(3) = 215$，解得$a = 8$。

M1	$10a + 45(3) = 215$，解得$a = 8$。
A1	结论、定义域、精度与所需单位正确

(c) · 2 marks

50.

由$a = 8, d = 3$，$u_{15} = 8 + 14(3) = 50$。

M1	由$a = 8, d = 3$，$u_{15} = 8 + 14(3) = 50$。
A1	结论、定义域、精度与所需单位正确

(d) · 2 marks

815.

$u_{21} = 68, u_{30} = 95$，十项和为$10(68 + 95)/2 = 815$。

M1	$u_{21} = 68, u_{30} = 95$，十项和为$10(68 + 95)/2 = 815$。
A1	结论、定义域、精度与所需单位正确

Answer 4

P2-C05-R04

原创改编（2026）· 主要考点：抽取等差子数列
考法参照：P2-2023-10-Q08（2023）：剧院排数等差模型
以索引抽取替代座位排列，考察子数列索引而非仅换情境数字。

(a) · 2 marks

$a = 3, d = 4.$

$5d = 20$，故$d = 4$；$a + 2d = 11$给$a = 3$。

M1	$5d = 20$，故$d = 4$；$a + 2d = 11$给$a = 3$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$v_n = 12n - 5.$

选中项的原编号为$3n - 1$，故$v_n = 3 + (3n - 2)4 = 12n - 5$。

M1	选中项的原编号为$3n - 1$，故$v_n = 3 + (3n - 2)4 = 12n - 5$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$1751.$

$3n - 1 \le 50$给$n \le 17$。$v_1 = 7, v_{17} = 199$，和为$17(7 + 199)/2 = 1751$。

M1	$3n - 1 \le 50$给$n \le 17$。$v_1 = 7, v_{17} = 199$，和为$17(7 + 199)/2 = 1751$。
A2	结论、定义域、精度与所需单位正确

Answer 5

P2-C05-R05

原创改编（2026）· 主要考点：无穷和反推公比
考法参照：P2-2022-10-Q08（2022）：无穷和约束与有限等比和
用有限和与余项反推公比，并明确剩余和而非累计阈值。

(a) · 2 marks

$r^3 = \dfrac{8}{27}$.

$S_3 = S_\infty(1 - r^3)$，所以$38 = 54(1 - r^3)$。

M1	$S_3 = S_\infty(1 - r^3)$，所以$38 = 54(1 - r^3)$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$a = 18$.

把已求公比代入无穷和公式，反求首项：$r = \dfrac{2}{3}$，$a = 54(1 - \dfrac{2}{3}) = 18$。

M1	把已求公比代入无穷和公式，反求首项：$r = \dfrac{2}{3}$，$a = 54(1 - \dfrac{2}{3}) = 18$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$n = 10$.

余项和为$54(\dfrac{2}{3})^n$；取对数得$n > \log(\dfrac{1}{54}) / \log(\dfrac{2}{3}) \approx 9.84$，所以$n = 10$。

M1	余项和为$54(\dfrac{2}{3})^n$；取对数得$n > \log(\dfrac{1}{54}) / \log(\dfrac{2}{3}) \approx 9.84$，所以$n = 10$。
A2	结论、定义域、精度与所需单位正确

Answer 6

P2-C05-R06

原创改编（2026）· 主要考点：相邻项和识别等比
考法参照：P2-2023-06-Q06（2023）：三项等比及首次超过总和
相邻项和辨识后抽取偶数项无限子级数，改变原题目标。

(a) · 2 marks

$r = \dfrac{1}{2}$.

$ar + ar^2 = 30$，下一相邻项和为前者乘r，故$r = \dfrac{15}{30}$。

M1	$ar + ar^2 = 30$，下一相邻项和为前者乘r，故$r = \dfrac{15}{30}$。
A1	结论、定义域、精度与所需单位正确

(b) Using your common ratio, find

(i) · 2 marks

40.

把公比代入第二、三项之和的条件：$a(\dfrac{1}{2} + \dfrac{1}{4}) = 30$。

M1	把公比代入第二、三项之和的条件：$a(\dfrac{1}{2} + \dfrac{1}{4}) = 30$。
A1	结论、定义域、精度与所需单位正确

(ii) · 2 marks

80.

公比绝对值小于1，代入无穷和公式：$|r| < 1$，$40/(1 - \dfrac{1}{2}) = 80$。

M1	公比绝对值小于1，代入无穷和公式：$|r| < 1$，$40/(1 - \dfrac{1}{2}) = 80$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$\dfrac{80}{3}$.

偶数项为$20, 5, \dfrac{5}{4}, \ldots$，公比$\dfrac{1}{4}$，和为$20/(1 - \dfrac{1}{4}) = \dfrac{80}{3}$。

M1	偶数项为$20, 5, \dfrac{5}{4}, \ldots$，公比$\dfrac{1}{4}$，和为$20/(1 - \dfrac{1}{4}) = \dfrac{80}{3}$。
A2	结论、定义域、精度与所需单位正确

Answer 7

P2-C05-R07

原创改编（2026）· 主要考点：负公比与余项符号
考法参照：P2-2022-06-Q06（2022）：负公比等比级数
增加奇偶误差和精度要求，不照搬原卷求指定有限和。

(a) · 2 marks

$S_\infty = \dfrac{8}{3}$, $S_n = \dfrac{8}{3}[1-(-\dfrac{1}{2})^n]$.

使用等比有限和及无穷和公式；$|r|<1$。

M1	使用等比有限和及无穷和公式；$\lvert r\rvert<1$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

奇数n时较大。

$(-\dfrac{1}{2})^n<0$，故$S_n=\dfrac{8}{3}[1+2^{-n}]>\dfrac{8}{3}$。

M1	$(-\dfrac{1}{2})^n<0$，故$S_n=\dfrac{8}{3}[1+2^{-n}]>\dfrac{8}{3}$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$n=9$.

误差为$8/(3\cdot 2^n)$；$n=8$约0.01042，$n=9$约0.00521。

M1	误差为$8/(3\cdot 2^n)$；$n=8$约0.01042，$n=9$约0.00521。
A2	结论、定义域、精度与所需单位正确

Answer 8

原创改编（2026）· 主要考点：等差项构成等比
考法参照：P2-2023-10-Q10（2023）：二项式项构成等比与三角等比级数
把二项式等比联系改为非等距抽取等差项，需排除零公差。

(a) · 3 marks

$d = 3a$.

$(a+3d)^2 = (a+d)(a+8d)$给$d^2 - 3ad = 0$，因$d \neq 0$可约去d。

M1	$(a+3d)^2 = (a+d)(a+8d)$给$d^2 - 3ad = 0$，因$d \neq 0$可约去d。
A2	结论、定义域、精度与所需单位正确

(b) · 2 marks

$r = \dfrac{5}{2}$.

$a + 4d = 13$及$d = 3a$给$a = 1, d = 3$，三项为$4, 10, 25$。

M1	$a + 4d = 13$及$d = 3a$给$a = 1, d = 3$，三项为$4, 10, 25$。
A1	结论、定义域、精度与所需单位正确

(c) · 2 marks

210.

把首项和公差代入等差级数求和公式：$S_{12} = 6[2 + 11(3)] = 210$。

M1	把首项和公差代入等差级数求和公式：$S_{12} = 6[2 + 11(3)] = 210$。
A1	结论、定义域、精度与所需单位正确

Answer 9

P2-C05-R09

原创改编（2026）· 主要考点：周期递推与求和
考法参照：P2-2023-10-Q02（2023）：周期递推数列求和
新二周期递推、奇数长度总和和错误收敛主张构成递进。

(a) · 2 marks

9, 3, 9.

逐次代入递推式。

M1	逐次代入递推式。
A1	结论、定义域、精度与所需单位正确

(b) Using the repeating pattern, find

(i) · 2 marks

3.

奇数位置的项恒为3。

M1	奇数位置的项恒为3。
A1	结论、定义域、精度与所需单位正确

(ii) · 2 marks

603.

50对各和12，再加3。

M1	50对各和12，再加3。
A1	结论、定义域、精度与所需单位正确

(c) · 2 marks

数列不收敛。

奇数项恒为3，偶数项恒为9，不会趋近同一数。

M1	奇数项恒为3，偶数项恒为9，不会趋近同一数。
A1	结论、定义域、精度与所需单位正确

Answer 10

P2-C05-R10

原创改编（2026）· 主要考点：平移递推成等比
考法参照：P2-2021-01-Q08（2021）：递推数列与前三项和
不问前三项的参数，改用平移数列建立通项及任意部分和。

(a) · 2 marks

$v_{n+1} = 3v_n$.

给递推两边同时加2，再提出公因子3：$u_{n+1}+2 = 3u_n+6 = 3(u_n+2)$。

M1	给递推两边同时加2，再提出公因子3：$u_{n+1}+2 = 3u_n+6 = 3(u_n+2)$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$u_n = 4\cdot 3^{n-1} - 2$.

$v_1 = 4$且公比3，所以$v_n = 4\cdot 3^{n-1}$，再减2。

M1	$v_1 = 4$且公比3，所以$v_n = 4\cdot 3^{n-1}$，再减2。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$S_n = 2(3^n - 1) - 2n$.

对$4\cdot 3^{k-1}$求等比和得$2(3^n-1)$，常数项总和为$-2n$。

M1	对$4\cdot 3^{k-1}$求等比和得$2(3^n-1)$，常数项总和为$-2n$。
A2	结论、定义域、精度与所需单位正确

Answer 11

P2-C05-R11

原创改编（2026）· 主要考点：递推模型与接近极限
考法参照：P2-2021-01-Q10（2021）：等差与等比训练模型比较
把纯训练等差等比比较改为有顺序含义的排液加液递推与极限。

(a) · 2 marks

$u_{n+1} = 0.8u_n + 6,\ u_1 = 10.$

先留下80%，再增加6升。

M1	先留下80%，再增加6升。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

30升。

$0.8^{n-1} \to 0$，故体积趋近30升。

M1	$0.8^{n-1} \to 0$，故体积趋近30升。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

第15次。

$0.8^{n-1} \le 0.05$给$n - 1 \ge 13.425\ldots$，故$n = 15$。

M1	$0.8^{n-1} \le 0.05$给$n - 1 \ge 13.425\ldots$，故$n = 15$。
A2	结论、定义域、精度与所需单位正确

Answer 12

原创改编（2026）· 主要考点：分段储蓄模型
考法参照：P2-2022-01-Q08（2022）：前等差后等比的累计距离
改变分段模型方向，后段衰减且要求无限总额，关注衔接项。

(a) · 2 marks

£460。

第8周为75，等差和为$8(40+75)/2=460$。

M1	第8周为75，等差和为$8(40+75)/2=460$。
A1	结论、定义域、精度与所需单位正确

(b) · 3 marks

£384.43（约）。

这8项首项67.5、公比0.9，和$67.5(1-0.9^8)/0.1\approx 384.43$。

M1	这8项首项67.5、公比0.9，和$67.5(1-0.9^8)/0.1\approx 384.43$。
A2	结论、定义域、精度与所需单位正确

(c) · 2 marks

£1135。

第9周起无穷和$67.5/(1-0.9)=675$，加前8周460。

M1	第9周起无穷和$67.5/(1-0.9)=675$，加前8周460。
A1	结论、定义域、精度与所需单位正确

Answer 13

P2-C05-R13

原创改编（2026）· 主要考点：两种递增方案比较
考法参照：P2-2021-01-Q10（2021）：等差与等比训练模型比较
保留比较类型，改成累计与单期阈值不同判断，模型数据及设问链均变。

(a) Find the distance in week 6 under

(i) · 2 marks

4 km。

等差模型的第6周等于首周加五次增量：$2+5(0.4)=4$。

M1	等差模型的第6周等于首周加五次增量：$2+5(0.4)=4$。
A1	结论、定义域、精度与所需单位正确

(ii) · 2 marks

约3.221 km。

等比模型的第6周已经历五次增长：$2(1.1)^5\approx 3.221$。

M1	等比模型的第6周已经历五次增长：$2(1.1)^5\approx 3.221$。
A1	结论、定义域、精度与所需单位正确

(b) · 3 marks

A：38 km；B：约31.875 km。

A和为$10[4+9(0.4)]/2$；B和为$2(1.1^{10}-1)/0.1$。

M1	A和为$10[4+9(0.4)]/2$；B和为$2(1.1^{10}-1)/0.1$。
A2	结论、定义域、精度与所需单位正确

(c) · 2 marks

第13周。

$2(1.1)^{n-1}>6$给$n>1+\log 3/\log 1.1=12.526\ldots$。

M1	$2(1.1)^{n-1}>6$给$n>1+\log 3/\log 1.1=12.526\ldots$。
A1	结论、定义域、精度与所需单位正确

Answer 14

P2-C05-R14

原创改编（2026）· 主要考点：等差和与前后段关系
考法参照：P2-2020-10-Q05（2020）：两种等差储蓄方案的合理根
加入相邻等长区段和，要求推导n的二次式，不是替换储蓄数据。

(a) · 2 marks

$n^2d = 200.$

后n项每项比对应前n项多nd，总差为$n^2d = 350 - 150$。

M1	后n项每项比对应前n项多nd，总差为$n^2d = 350 - 150$。
A1	结论、定义域、精度与所需单位正确

(b) · 3 marks

$3n^2 - 25n - 50 = 0.$

$150 = \frac{n}{2}[12 + (n-1)d]$；代$d = 200/n^2$得$150 = 6n + 100 - 100/n$，乘$n$整理并除2。

M1	$150 = \frac{n}{2}[12 + (n-1)d]$；代$d = 200/n^2$得$150 = 6n + 100 - 100/n$，乘$n$整理并除2。
A2	结论、定义域、精度与所需单位正确

(c) · 2 marks

$n = 10, d = 2.$

$3n^2 - 25n - 50 = (n-10)(3n+5) = 0$，只取正整数$n = 10$，再得$d = 2$。

M1	$3n^2 - 25n - 50 = (n-10)(3n+5) = 0$，只取正整数$n = 10$，再得$d = 2$。
A1	结论、定义域、精度与所需单位正确

Answer 15

原创改编（2026）· 主要考点：等比奇偶子级数
考法参照：P2-2022-10-Q08（2022）：无穷和约束与有限等比和
用正负子级数重建负公比，再连有限和。

(a) · 2 marks

$r = -\dfrac{1}{3}$.

偶数子级数为奇数子级数乘r，所以$r = -\dfrac{4}{12}$。

M1	偶数子级数为奇数子级数乘r，所以$r = -\dfrac{4}{12}$。
A1	结论、定义域、精度与所需单位正确

(b) For the geometric series determined in part (a), find

(i) · 2 marks

$\dfrac{32}{3}$.

$a/(1-r^2) = 12$，代$r = -\dfrac{1}{3}$。

M1	$a/(1-r^2) = 12$，代$r = -\dfrac{1}{3}$。
A1	结论、定义域、精度与所需单位正确

(ii) · 3 marks

$\dfrac{640}{81}$.

总和$12 - 4 = 8$，$S_4 = 8(1 - \dfrac{1}{81})$。

M1	总和$12 - 4 = 8$，$S_4 = 8(1 - \dfrac{1}{81})$。
A2	结论、定义域、精度与所需单位正确

Answer 16

P2-C05-R16

原创改编（2026）· 主要考点：叠层图案的总量
考法参照：P2-2023-10-Q08（2023）：剧院排数等差模型
以明确新图案、材料整数限制和未用量组织应用题。

(a) · 2 marks

$4n+3$.

等差通项$7+4(n-1)$。

M1	等差通项$7+4(n-1)$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$S_n=n(2n+5)$.

用等差级数公式，将首项7和公差4代入并化简：$S_n=\frac{n}{2}[14+4(n-1)]$。

M1	用等差级数公式，将首项7和公差4代入并化简：$S_n=\frac{n}{2}[14+4(n-1)]$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

13行，剩15块。

$S_{13}=403$，$S_{14}=462>418$；余数$418-403=15$。

M1	$S_{13}=403$，$S_{14}=462>418$；余数$418-403=15$。
A2	结论、定义域、精度与所需单位正确

Answer 17

P2-C05-R17

原创改编（2026）· 主要考点：反弹距离级数
考法参照：P2-2022-10-Q08（2022）：无穷和约束与有限等比和
加入初始单程与后续双程的路径几何，避免无限和直接套公式。

(a) · 2 marks

1.08 m。

反弹高度为$5(0.6)^n$，第三次为$5(0.6)^3$。

M1	反弹高度为$5(0.6)^n$，第三次为$5(0.6)^3$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

11 m。

首次落下5米，第一次反弹上、下各3米，所以$5+2(3)=11$。

M1	首次落下5米，第一次反弹上、下各3米，所以$5+2(3)=11$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

20 m。

$5+2[3/(1-0.6)]=20$；首落5米只计一次。

M1	$5+2[3/(1-0.6)]=20$；首落5米只计一次。
A2	结论、定义域、精度与所需单位正确

Answer 18

原创改编（2026）· 主要考点：等比缩放图形

考法参照：P2-2022-10-Q08（2022）：无穷和约束与有限等比和

用边长比例与面积比例差异形成新图形关联及阈值。

(a) · 2 marks

$A_n = 81(\frac{4}{9})^{n-1}\ \text{cm}^2$.

长度缩为$\frac{2}{3}$，面积缩为$(\frac{2}{3})^2 = \frac{4}{9}$。

M1	长度缩为$\frac{2}{3}$，面积缩为$(\frac{2}{3})^2 = \frac{4}{9}$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$\frac{729}{5}\ \text{cm}^2$.

无限等比和$81/(1-\frac{4}{9}) = \frac{729}{5}$。

M1	无限等比和$81/(1-\frac{4}{9}) = \frac{729}{5}$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

第7个。

$81(\frac{4}{9})^{n-1} < 1$给$n-1 > 5.419\ldots$，故$n = 7$。

M1	$81(\frac{4}{9})^{n-1} < 1$给$n-1 > 5.419\ldots$，故$n = 7$。
A2	结论、定义域、精度与所需单位正确

Answer 19

P2-C05-R19

原创改编（2026）· 主要考点：交错通项与区段求和
考法参照：P2-2020-01-Q08（2020）：等差级数证明与交错项求和
引入偶数部分和证明与非首项区段求和，避免原卷奇数总和任务。

(a) · 2 marks

$7, 1, 11$.

分别代$n = 1, 2, 3$，注意交错符号。

M1	分别代$n = 1, 2, 3$，注意交错符号。
A1	结论、定义域、精度与所需单位正确

(b) · 3 marks

$S_{2m} = 4m(m+1)$.

交错项成对抵消，$\sum_{n=1}^{2m}(2n+1) = 2m(2m+1) + 2m$。

M1	交错项成对抵消，$\sum_{n=1}^{2m}(2n+1) = 2m(2m+1) + 2m$。
A2	结论、定义域、精度与所需单位正确

(c) · 2 marks

840.

所求区段和等于前30项和减去前10项和：$S_{30} - S_{10} = 4(15)(16) - 4(5)(6) = 960 - 120 = 840$。

M1	所求区段和等于前30项和减去前10项和：$S_{30} - S_{10} = 4(15)(16) - 4(5)(6) = 960 - 120 = 840$。
A1	结论、定义域、精度与所需单位正确

Answer 20

P2-C05-R20

原创改编（2026）· 主要考点：周期三角数列
考法参照：P2-2022-10-Q03（2022）：周期数列与等差求和结合
不同周期函数连接线性序列差，改变周期及求和结构。

(a) · 2 marks

$2, 1, 2, 1$.

$\sin^2(n\pi/2)$依次为$1, 0, 1, 0$。

M1	$\sin^2(n\pi/2)$依次为$1, 0, 1, 0$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

113.

37对各和3，另加第75项2，得$37(3)+2=113$。

M1	37对各和3，另加第75项2，得$37(3)+2=113$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

8437.

对线性项用等差求和，再减去上一问得到的周期部分和：$3\cdot 75\cdot 76/2-113=8550-113=8437$。

M1	对线性项用等差求和，再减去上一问得到的周期部分和：$3\cdot 75\cdot 76/2-113=8550-113=8437$。
A2	结论、定义域、精度与所需单位正确

Answer 21

P2-C05-R21

原创改编（2026）· 主要考点：递推分式周期
考法参照：P2-2023-10-Q02（2023）：周期递推数列求和
选用有定义的三周期有理递推，要求解释周期及整组余项。

(a) · 2 marks

$\frac{1}{2}, -1, 2.$

逐项代入，可见返回初值。

M1	逐项代入，可见返回初值。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

每3项重复，$u_{100} = 2$。

递推只依赖当前项，$u_4 = u_1$后循环重复；$100 \equiv 1 \pmod 3$。

M1	递推只依赖当前项，$u_4 = u_1$后循环重复；$100 \equiv 1 \pmod 3$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$\frac{103}{2}.$

每组三项和$2 + \frac{1}{2} - 1 = \frac{3}{2}$，33组再加2，得$\frac{103}{2}$。

M1	每组三项和$2 + \frac{1}{2} - 1 = \frac{3}{2}$，33组再加2，得$\frac{103}{2}$。
A2	结论、定义域、精度与所需单位正确

Answer 22

原创改编（2026）· 主要考点：通项与部分和之差
考法参照：P2-2021-10-Q02（2021）：递推与前三项和
从混合部分和重建通项及递推，非给定递推求前三项。

(a) · 2 marks

7.

首项就是只含一项的部分和：$S_1 = 4(2-1)+3 = 7$。

M1	首项就是只含一项的部分和：$S_1 = 4(2-1)+3 = 7$。
A1	结论、定义域、精度与所需单位正确

(b) · 3 marks

$u_n = 2^{n+1} + 3$.

$n \geq 2$时作差$S_n - S_{n-1} = 4(2^n - 2^{n-1}) + 3$；$n = 1$也吻合。

M1	$n \geq 2$时作差$S_n - S_{n-1} = 4(2^n - 2^{n-1}) + 3$；$n = 1$也吻合。
A2	结论、定义域、精度与所需单位正确

(c) · 2 marks

$u_{n+1} = 2u_n - 3$.

把通项代入递推右边，整理后恰好为下一项通项：$2(2^{n+1}+3) - 3 = 2^{n+2} + 3 = u_{n+1}$。

M1	把通项代入递推右边，整理后恰好为下一项通项：$2(2^{n+1}+3) - 3 = 2^{n+2} + 3 = u_{n+1}$。
A1	结论、定义域、精度与所需单位正确

Answer 23

P2-C05-R23

原创改编（2026）· 主要考点：指数生产模型的整数阈值
考法参照：P2-2023-01-Q07（2023）：等比级数超过给定和
区分单期与累计、加入末段占比，增长关系非原卷复用。

(a) · 2 marks

$S_n = 2400(1.05^n - 1).$

等比和$120(1.05^n-1)/(1.05-1)$。

M1	等比和$120(1.05^n-1)/(1.05-1)$。
A1	结论、定义域、精度与所需单位正确

(b) · 3 marks

$n = 13.$

由总产量不等式分离指数项，再取对数；最后向上取整数：$1.05^n > \dfrac{11}{6}$，$n > \log(\dfrac{11}{6})/\log 1.05 \approx 12.42$。

M1	由总产量不等式分离指数项，再取对数；最后向上取整数：$1.05^n > \dfrac{11}{6}$，$n > \log(\dfrac{11}{6})/\log 1.05 \approx 12.42$。
A2	结论、定义域、精度与所需单位正确

(c) · 2 marks

约0.3073，即30.7%。

比例为$(S_{12}-S_9)/S_{12} = (1.05^{12}-1.05^9)/(1.05^{12}-1) \approx 0.3072516$。

M1	比例为$(S_{12}-S_9)/S_{12} = (1.05^{12}-1.05^9)/(1.05^{12}-1) \approx 0.3072516$。
A1	结论、定义域、精度与所需单位正确

Answer 24

P2-C05-R24

原创改编（2026）· 主要考点：分期支出等差模型
考法参照：P2-2021-10-Q05（2021）：逐年减少产量与累计总和
有限有效期、非正付款停止和百分比累计阈值构成新约束。

(a) · 2 marks

15次。

$900-60(n-1)>0$给$n<16$，最大$n=15$。

M1	$900-60(n-1)>0$给$n<16$，最大$n=15$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

£7200。

最后付款60，$15(900+60)/2=7200$。

M1	最后付款60，$15(900+60)/2=7200$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

8年。

目标5400，$S_n=30n(31-n)$；$S_7=5040$，$S_8=5520$。

M1	目标5400，$S_n=30n(31-n)$；$S_7=5040$，$S_8=5520$。
A2	结论、定义域、精度与所需单位正确

Answer 25

P2-C05-R25

原创改编（2026）· 主要考点：两个连续部分和
考法参照：P2-2023-06-Q06（2023）：三项等比及首次超过总和
改为从N与N+1部分和差识别相邻项，增加后半段和。

(a) · 2 marks

$r = 2$.

两差分别为相邻项u_{n+1}, u_n，比值为2。

M1	两差分别为相邻项u_{n+1}, u_n，比值为2。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$n = 5$.

$u_n = 6 \cdot 2^{n-1} = 96$给$2^{n-1} = 16$。

M1	$u_n = 6 \cdot 2^{n-1} = 96$给$2^{n-1} = 16$。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

5952.

利用已求的项数，将两个等比部分和作差：$S_{10} - S_5 = 6(2^{10} - 2^5) = 5952$。

M1	利用已求的项数，将两个等比部分和作差：$S_{10} - S_5 = 6(2^{10} - 2^5) = 5952$。
A2	结论、定义域、精度与所需单位正确

Answer 26

P2-C05-R26

原创改编（2026）· 主要考点：等比级数公式证明

考法参照：P2-2020-10-Q08（2020）：等比求和公式证明及负公比

保留经典求和证明但后续改为区间和与尾和，避免同一数列特定项换数。

(a) · 3 marks

$S_n = \dfrac{a(1-r^n)}{1-r}$.

$S_n - rS_n = a - ar^n$，两边除以$1-r$。

M1	$S_n - rS_n = a - ar^n$，两边除以$1-r$。
A2	结论、定义域、精度与所需单位正确

(b) A geometric series has $a = 9$ and $r = \dfrac{1}{3}$. Find

(i) · 2 marks

$\dfrac{40}{27}$.

第三项1，四项和$(1-\frac{1}{81})/(1-\frac{1}{3})$。

M1	第三项1，四项和$(1-\frac{1}{81})/(1-\frac{1}{3})$。
A1	结论、定义域、精度与所需单位正确

(ii) · 2 marks

$\dfrac{1}{54}$.

第7项$\frac{1}{81}$，无限尾和$(\frac{1}{81})/(1-\frac{1}{3})$。

M1	第7项$\frac{1}{81}$，无限尾和$(\frac{1}{81})/(1-\frac{1}{3})$。
A1	结论、定义域、精度与所需单位正确

Answer 27

原创改编（2026）· 主要考点：等差中点与平方和约束
考法参照：P2-2023-06-Q06（2023）：三项等比及首次超过总和
以等差平方约束连接等比，不是原卷给定三个代数等比项。

(a) · 2 marks

$d^2 = 16$.

平方和$300 + 2d^2 = 332$。

M1	平方和$300 + 2d^2 = 332$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$6, 10, 14$.

取$d = 4$给递增顺序。

M1	取$d = 4$给递增顺序。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

$\dfrac{4802}{27}$.

先由指定的前两项求公比，再用等比通项计算第五项：$a = 6, r = \dfrac{14}{6} = \dfrac{7}{3}$，$u_5 = 6(\dfrac{7}{3})^4 = \dfrac{4802}{27}$。

M1	先由指定的前两项求公比，再用等比通项计算第五项：$a = 6, r = \dfrac{14}{6} = \dfrac{7}{3}$，$u_5 = 6(\dfrac{7}{3})^4 = \dfrac{4802}{27}$。
A2	结论、定义域、精度与所需单位正确

Answer 28

P2-C05-R28

原创改编（2026）· 主要考点：周期费用与区段和
考法参照：P2-2022-10-Q03（2022）：周期数列与等差求和结合
周期级数应用带一次性费用及奇偶预算边界，不沿用三角周期原式。

(a) · 2 marks

184单位。

13个奇数周期及12个偶数周期：$20+13(8)+12(5)=184$。

M1	13个奇数周期及12个偶数周期：$20+13(8)+12(5)=184$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$20+13n$.

每对周期13单位，共n对，再加一次启动。

M1	每对周期13单位，共n对，再加一次启动。
A1	结论、定义域、精度与所需单位正确

(c) · 3 marks

73个。

72个用$20+36(13)=488$；73个用496；74个用501，故73。

M1	72个用$20+36(13)=488$；73个用496；74个用501，故73。
A2	结论、定义域、精度与所需单位正确

Answer 29

P2-C05-R29

原创改编（2026）· 主要考点：等比尾和、公比正负、N与N+2关系

考法参照：P2-2022-10-Q08（2022）：无穷和约束与有限等比和

从相隔两项的尾和比先得正负公比，利用未知整数索引与尾和符号排除负支，再求N到2N区段和；不沿用先求r再求阈值的旧结构。

(a) · 2 marks

$r = \pm\dfrac{1}{2}.$

第N项后的尾和为$96r^N$；再过两项尾和乘r^2，所以$r^2 = \dfrac{3}{12} = \dfrac{1}{4}$。

M1	第N项后的尾和为$96r^N$；再过两项尾和乘r^2，所以$r^2 = \dfrac{3}{12} = \dfrac{1}{4}$。
A1	结论、定义域、精度与所需单位正确

(b) · 3 marks

$r = \dfrac{1}{2},\ N = 3.$

$r^N = \dfrac{12}{96} = \dfrac{1}{8}$。绝对值得$2^{-N} = \dfrac{1}{8}$故$N = 3$为奇数；若$r = -\dfrac{1}{2}$则$r^N < 0$，与正尾和矛盾。

M1	$r^N = \dfrac{12}{96} = \dfrac{1}{8}$。绝对值得$2^{-N} = \dfrac{1}{8}$故$N = 3$为奇数；若$r = -\dfrac{1}{2}$则$r^N < 0$，与正尾和矛盾。
A2	结论、定义域、精度与所需单位正确

(c) · 3 marks

$a = 48$；区段和$\dfrac{21}{2}.$

$a = 96(1-r) = 48$；所求为第4至第6项，等于尾和$12 - 96(\dfrac{1}{2})^6 = 12 - \dfrac{3}{2} = \dfrac{21}{2}$。

M1	$a = 96(1-r) = 48$；所求为第4至第6项，等于尾和$12 - 96(\dfrac{1}{2})^6 = 12 - \dfrac{3}{2} = \dfrac{21}{2}$。
A2	结论、定义域、精度与所需单位正确

Answer 30

P2-C05-R30

原创改编（2026）· 主要考点：坐标图与等差增长
考法参照：P2-2020-01-Q08（2020）：等差级数证明与交错项求和
以离散坐标线重建通项并考N与N+1和差，提供新图。

(a) · 2 marks

$u_n = 3n + 3.$

斜率$(24-9)/(7-2) = 3$，截距$9 - 3(2) = 3$。

M1	斜率$(24-9)/(7-2) = 3$，截距$9 - 3(2) = 3$。
A1	结论、定义域、精度与所需单位正确

(b) · 2 marks

$S_n = \dfrac{3}{2}n(n+3).$

$u_1 = 6, d = 3$，代等差和公式。

M1	$u_1 = 6, d = 3$，代等差和公式。
A1	结论、定义域、精度与所需单位正确

(c) · 2 marks

$n = 18.$

左边为$u_{n+1} = 3n + 6$，所以$n = 18$。

M1	左边为$u_{n+1} = 3n + 6$，所以$n = 18$。
A1	结论、定义域、精度与所需单位正确